

Software Engineering Lab - 1

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GitHub Link: <https://github.com/kaushik-dwarakanath/PES1UG24CS900-SE-Labs/tree/main/Lab1>

Requirements Specification

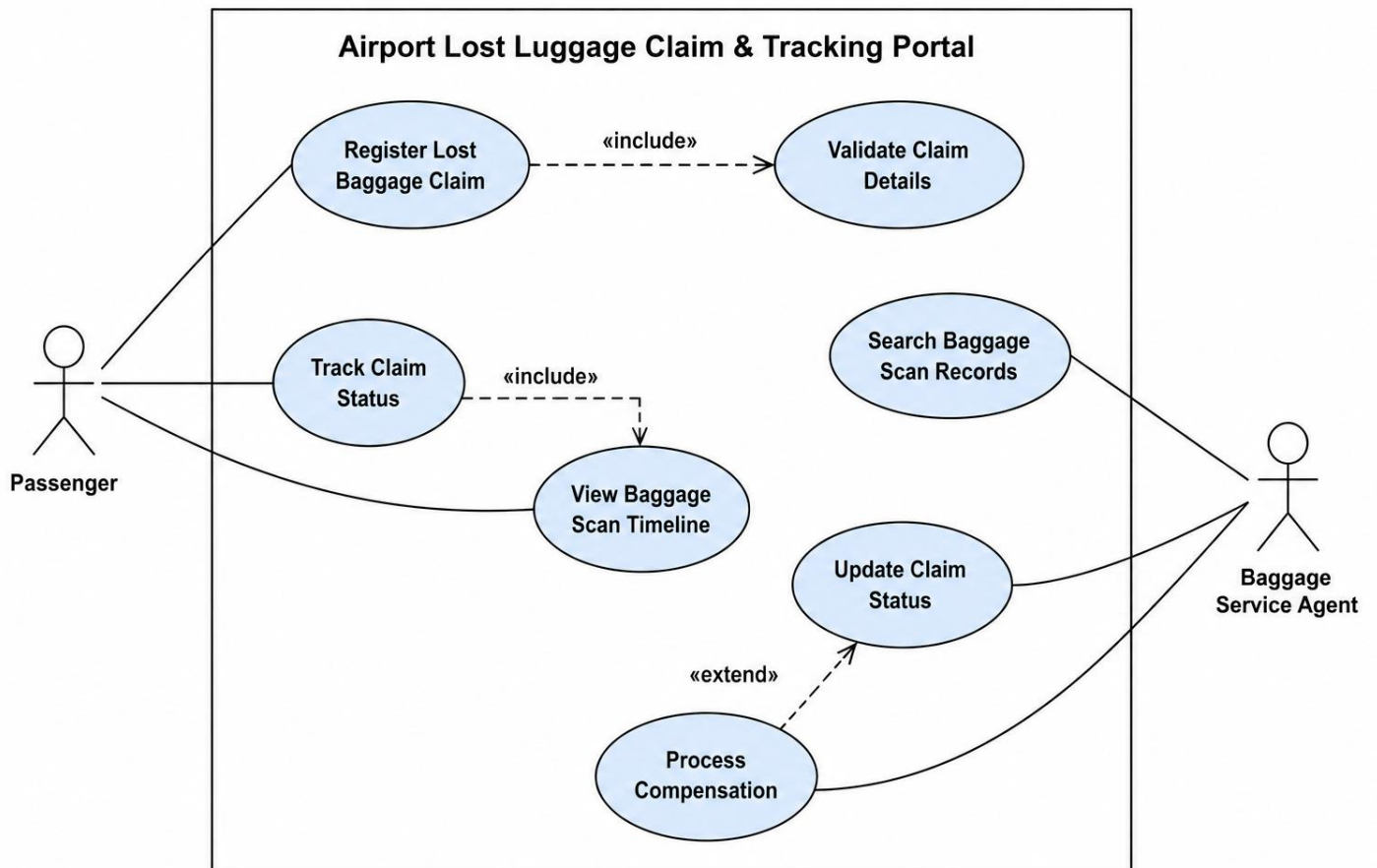
Functional Requirements

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall allow a passenger to register a lost baggage claim using a valid baggage tag number and flight details.	High	Pass: A claim is successfully created when valid baggage tag and flight details are provided. Fail: The system rejects incomplete or invalid information.	Enables passengers to officially report lost baggage.
FR-002	The system shall cross-reference the passenger's lost baggage claim with baggage scan records across connected airport terminals.	High	Pass: Matching baggage scan records are identified and linked to the claim. Fail: No incorrect baggage is matched.	Helps locate lost baggage accurately.
FR-003	The system shall allow passengers to track the current status of their baggage claim and view the baggage scan timeline.	High	Pass: The passenger can view the latest claim status and scan history. Fail: Incorrect or unavailable status is displayed.	Provides transparency during baggage recovery.
FR-004	The system shall allow a baggage service agent to update the claim status after locating, transferring, or delivering baggage.	Medium	Pass: Updated status is immediately reflected in the system. Fail: Status updates are not saved.	Keeps passengers informed of recovery progress.
FR-005	The system shall automatically initiate the passenger compensation process if baggage is not recovered within the airline's compensation policy.	Medium	Pass: Compensation workflow is automatically triggered when eligibility conditions are met. Fail: Eligible claims are not processed.	Ensures timely compensation and reduces manual effort.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The baggage tracking system shall return the complete baggage scan timeline within 200 milliseconds under normal operating conditions.	High	Pass: Average response time is less than 200 ms during performance testing. Fail: Response time exceeds 200 ms.	Provides a fast and responsive user experience.
NFR-002	Security	Only authenticated passengers and authorized baggage service agents shall be allowed to access baggage claim information.	High	Pass: Unauthorized users cannot access or modify baggage claim data. Fail: Unauthorized access is possible.	Protects passenger information and maintains data confidentiality.

Use Case Diagram



Use Case Specification

Register Lost Baggage Claim

Goal

Allow a passenger to register a lost baggage claim so that the airline can initiate baggage tracking and recovery.

Primary Actor

- Passenger

Supporting Actor

- Baggage Service Agent

Trigger

The passenger realizes that their checked baggage has not arrived after the flight and decides to report it.

Preconditions

- 1 The passenger has completed their flight.
- 2 The passenger possesses a valid baggage tag number.
- 3 The passenger has a valid ticket or booking reference.
- 4 The Airport Lost Luggage Claim & Tracking Portal is operational.

Postconditions

Success

- A lost baggage claim is successfully registered.
- A unique Claim ID is generated.
- The baggage recovery process is initiated.
- The claim becomes available to baggage service agents.

Failure

- No claim is created.
- The passenger is informed about the reason for failure.

Main Success Scenario

- 1 The passenger opens the Airport Lost Luggage Claim & Tracking Portal.
- 2 The passenger selects **Register Lost Baggage Claim**.
- 3 The system prompts the passenger to enter:
 - Baggage Tag Number
 - Flight Number
 - Passenger Details
- 4 The passenger enters the required information.

- 5 The system validates the submitted details.
- 6 The system searches baggage scan records across connected airport terminals.
- 7 The system creates a new lost baggage claim.
- 8 The system generates a unique Claim ID.
- 9 The system displays a confirmation message.
- 10 The passenger can later use the Claim ID to track baggage status.

Alternate Flow

Invalid Baggage Tag Number

- 1 The passenger enters an invalid baggage tag number.
- 2 The system detects that the baggage tag cannot be verified.
- 3 The system displays an error message indicating that the baggage tag is invalid.
- 4 The passenger re-enters the correct baggage information.
- 5 The use case resumes from Step 5 of the Main Success Scenario.

Business Rules

- Every baggage claim must be associated with exactly one baggage tag.
- A Claim ID must be unique.
- Only authenticated baggage service agents may update the claim status.
- Compensation processing can begin only after baggage is declared unrecoverable according to airline policy.